

Hypothesis:

Slate is triggering some slow code paths even when nothing is changing on the screen, possibly linked with some internal invalidation.

Originally I suspected SContentBrowser was the worst offender but it could just be because it has the most internal widgets. I am less confident of this after further testing.

The 2nd biggest offender could be the Details Panel with an actor selected.

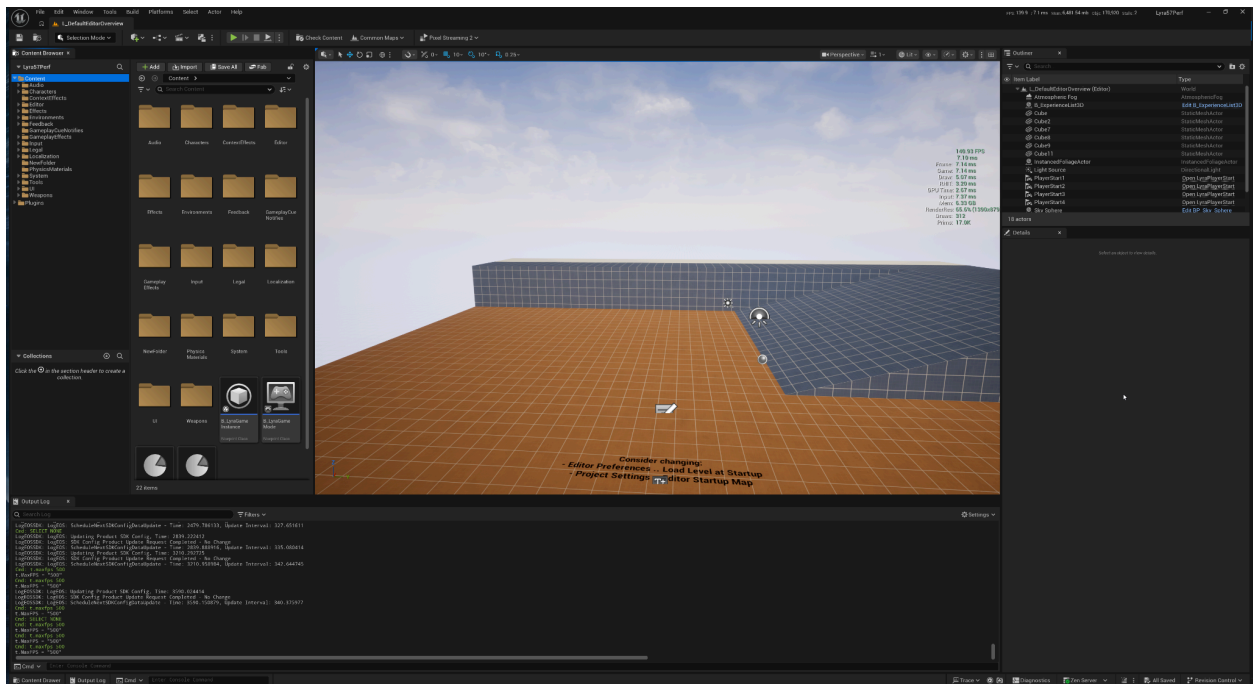
The setup:

Brand new Lyra 5.7 sample project on version UE 5.7.3 as I believe better mimics a potential real world project then a completely blank empty one.

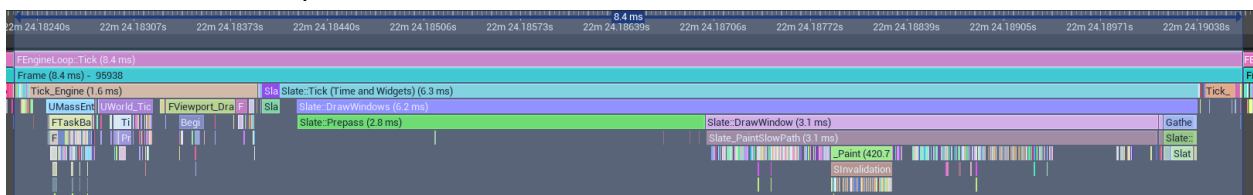
Console commands used

- stat unit
- stat fps
- t.maxfps 500 (since the default caps at 120 if left to 0 due to scalability settings)

What my screen generally looks like when testing.

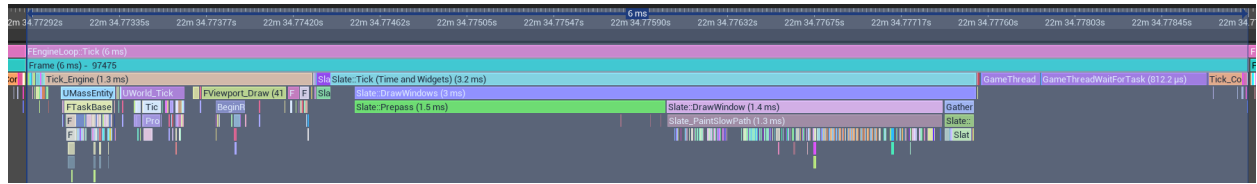


With content browser open

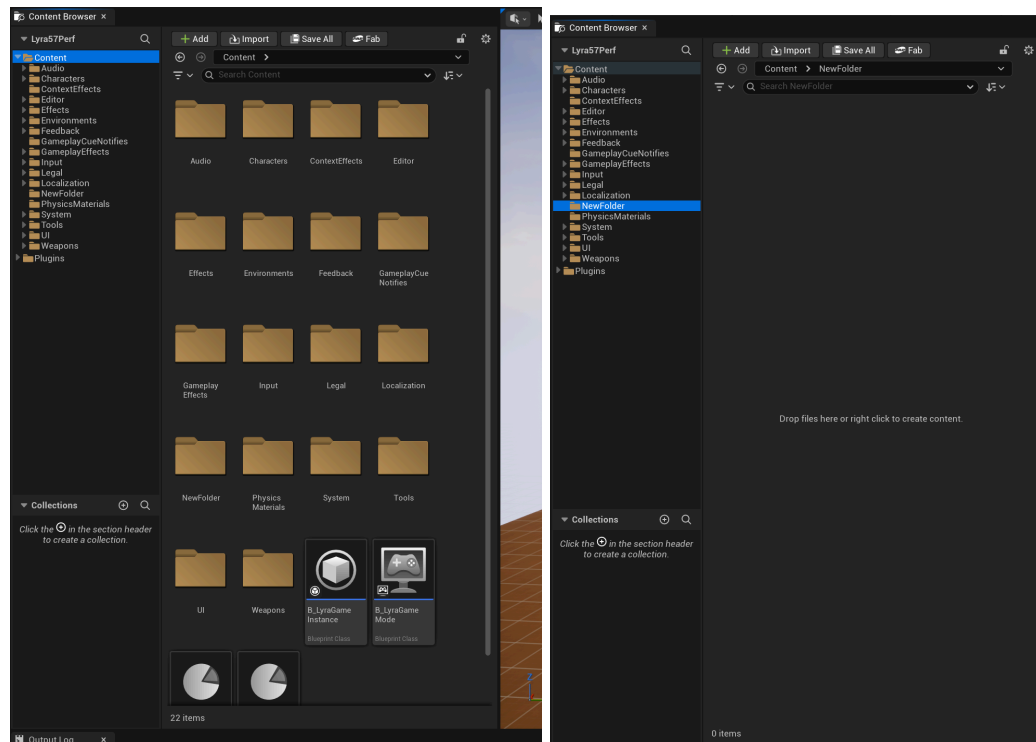


The main difference is nearly double `Slate::Tick`.
1/2 in prepass and 1/2 in `DrawWindow / PaintSlowPath`

With content browser open



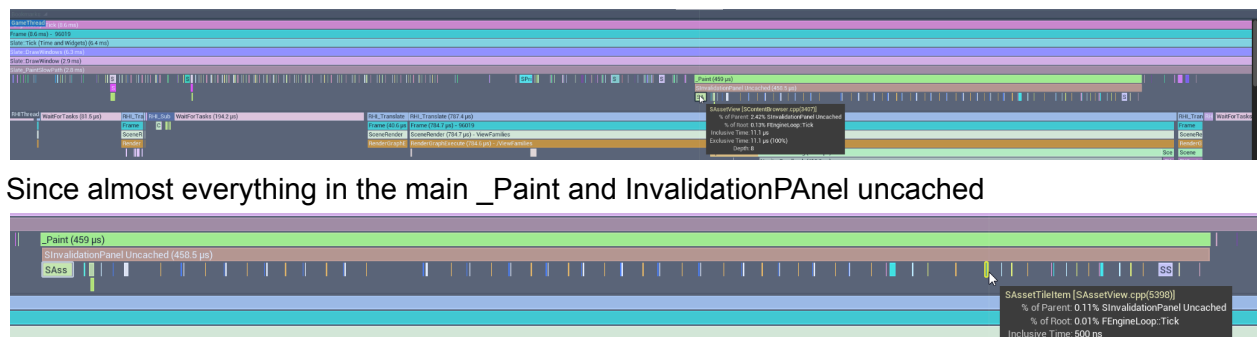
I was 1 of 2 views Default Content or clicked into an empty new folder



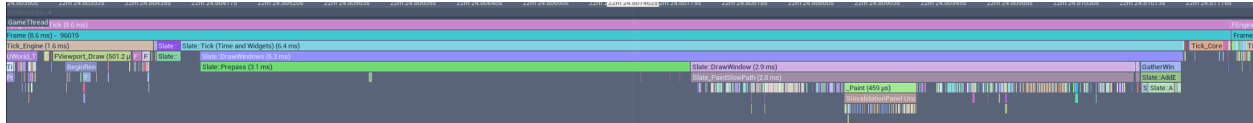
I did see some small changes from Details, Outliner, and Output log being open/not open but they appeared less of an impacts on the overall frame time.

Also used F10 and F11 to be in different PIE viewport expanded modes.

I initially was suspecting that something in the SContentBrowser was getting invalidated. Triggering the Slate_PaintSlowPath



Is from SContentBrowser and SAssetView and other things



```
bool bExecuteSlowPath = bNeedsSlowPath;
if (bExecuteSlowPath)
{
    SCOPED_NAMED_EVENT(Slate_PaintSlowPath, FColor::Red);
    CachedMaxLayerId = PaintSlowPath(Context);
}
```

I know Slate has some overhead but I still suspect something unnecessary might be getting invalidated.

Frame (36)	2	8.5 ms	53.2 µs
Slate::Tick (Time and Widgets) (1)	1	6.4 ms	126.5 µs
Slate::DrawWindows (3)	1	6.3 ms	33.4 µs
Slate::Prepass (2)	1	3.1 ms	3.1 ms
SPrioritizedWrapBox::ComputeDesiredSize (0)	1	18.5 µs	18.5 µs
SWidget::Invalidate (0)	3	900 ns	900 ns
Slate::DrawWindow (1)	2	2.9 ms	13.5 µs
Slate_PaintSlowPath (133)	2	2.9 ms	2.2 ms
GatherWindowElements (1)	2	254.1 µs	3.8 µs
Tick_Engine (13)	2	1.5 ms	224.1 µs
Tick_Core (2)	1	215.1 µs	209.9 µs

The main SlatePrepass has:

```
if (!GSlateIsOnFastUpdatePath || bNeedsPrepass)
```

- GSlateIsOnFastUpdatePath:
 - Whether or not we currently Painting/Updating the widget from the FastUpdate path (global invalidation).
- bNeedsPrepass
 - Are we currently updating the desired size?

I later added a breakpoint in the SWidget::Invalidate function.

If I add a breakpoint to that SWidget::Invalidate (after the early out of Reason::None). It appears some internal mechanism is constantly bouncing between these 2 every render frame.

```
// and it seems like it's toggling between these 2 states

UnrealEditor-SlateCore.dll!SWidget::Invalidate(EInvalidateWidgetReason InvalidateReason) Line 1265 C++
UnrealEditor-SlateCore.dll!SWidget::AssignParentWidget(TSharedPtr<SWidget,1> InParent) Line 742 C++
UnrealEditor-SlateCore.dll!FSlotBase::AfterContentOrOwnerAssigned() Line 92 C++
[Inline Frame] UnrealEditor-Slate.dll!FSlotBase::AttachWidget(TSharedRef<SWidget,1> &&) Line 51 C++
[Inline Frame] UnrealEditor-Slate.dll!TSlotBase<SHorizontalBox>::FSlot::operator[](TSharedRef<SWidget,1> &&) Line 128 C++
UnrealEditor-Slate.dll!SWindowTitleBar::UpdateWindowMenu(TSharedPtr<SWidget,1> MenuContent) Line 61 C++
UnrealEditor-Slate.dll!FTabManager::UpdateMainMenu(TSharedPtr<SDockTab,1> ForTab, const bool bForce) Line 896 C++
UnrealEditor-Slate.dll!FGlobalTabManager::UpdateMainMenu(const TSharedRef<SDockTab,1> & ForTab, const bool bForce) Line 3482 C++
UnrealEditor-Slate.dll!SDockingArea::OnOwningWindowActivated() Line 1028 C++
[External Code]

and

UnrealEditor-SlateCore.dll!SWidget::Invalidate(EInvalidateWidgetReason InvalidateReason) Line 1265 C++
UnrealEditor-SlateCore.dll!SWidget::ConditionallyDetachParentWidget(SWidget * InExpectedParent) Line 768 C++
[Inline Frame] UnrealEditor-Slate.dll!FSlotBase::AttachWidget(TSharedRef<SWidget,1> &&) Line 49 C++
[Inline Frame] UnrealEditor-Slate.dll!TSlotBase<SHorizontalBox>::FSlot::operator[](TSharedRef<SWidget,1> &&) Line 128 C++
UnrealEditor-Slate.dll!SWindowTitleBar::UpdateWindowMenu(TSharedPtr<SWidget,1> MenuContent) Line 61 C++
UnrealEditor-Slate.dll!FTabManager::UpdateMainMenu(TSharedPtr<SDockTab,1> ForTab, const bool bForce) Line 896 C++
UnrealEditor-Slate.dll!FGlobalTabManager::UpdateMainMenu(const TSharedRef<SDockTab,1> & ForTab, const bool bForce) Line 3482 C++
UnrealEditor-Slate.dll!SDockingArea::OnOwningWindowActivated() Line 1028 C++
[External Code]
```

Something is constantly detaching and reassigning something somewhere.
The above 2 callstacks constantly trigger for me back and forth like a tennis tournament.

Anyway, maybe the main perf loss from ContentBrowser is just generic Slate overhead. And there is nothing to be done. Or maybe there is some accidental invalidation triggering some unnecessary recalculation/redraw slower paths

I explored some other potential breakpoints to see if they were triggering but found no other interesting leads.

I've currently exhausted all my potential leads on this investigation at this time.